

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Ans. RAM Reseating Steps

1. **Turn off the computer** – Shut down the PC/laptop and unplug the power cable.
2. **Open the case/back cover** – Carefully remove the side panel or laptop back cover.
3. **Find the RAM slot** – Locate the long RAM module on the motherboard.
4. **Remove the RAM** – Open the small clips on both sides and gently take out the RAM.
5. **Check the RAM** – Look for dust, dirt, or any visible damage.
6. **Re-insert the RAM** – Align the notch on the RAM with the slot and press it down until the clips lock.
7. **Close the cover** – Put the case/back cover back on and reconnect the power.
8. **Start the computer** – Turn it on and check that it boots normally.

Issues observed: No physical damage was seen, and the RAM was successfully re-inserted. The computer started normally after reseating the RAM.

**2.Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.**

Step	What to Check	What to Do
1. Check the symptom	No display, screen flickering, or game crashes.	Note exactly what problem is happening.
2. Check the monitor	Monitor and display cable.	Make sure the monitor is on and the cable is connected properly.
3. Check GPU connections	GPU and power cables.	Turn off the PC and make sure the GPU is firmly seated and power cables are connected.
4. Check the driver	GPU driver.	Update or reinstall the correct graphics driver.
5. Check GPU temperature	GPU getting too hot.	Make sure the GPU fans are working and clean any dust.
6. Test the GPU	Run a game or graphics test.	Check if the display and games work normally.
7. Confirm the fix	Original problem.	Restart the PC and test again. If the display is stable and games run normally, the problem is likely fixed.

3.Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Ans. Storage Issue Simulation – SATA/NVMe Drive

1. **Turn off the PC** and unplug the power cable.
2. **Disconnect the SSD/HDD** from the motherboard or SATA cable.
3. **Turn on the PC** and enter the BIOS/UEFI.
4. **Before reconnecting:** The BIOS may show “**No Storage Device Detected**” or the drive name may be missing. Windows may also show “**No boot device found.**”
5. **Turn off the PC again** and reconnect the SSD/HDD firmly.
6. **Start the PC and check the BIOS:** The drive should appear again under the storage or boot-device list.
7. **After reconnecting:** Windows should start normally if the drive and connections are working.

Example observation: Before reconnecting, the SSD was not shown in BIOS and the PC could not find the boot drive. After reconnecting it, the SSD appeared in BIOS and the PC booted normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Ans. PC Randomly Restarts – Troubleshooting

Component I would check first: RAM

Reason: Faulty or loose RAM can cause random restarts, freezing, or crashes. It is also easy to check and reseat.

Diagnostic Steps

1. **Turn off the PC** and unplug the power.
2. **Open the case** and find the RAM sticks.
3. **Remove and reinsert the RAM** firmly into the slot.
4. **Check for dust or damage** on the RAM and slots.
5. **Test one RAM stick at a time** if there are multiple sticks.
6. **Start the PC and use it for some time** to see if the random restarting continues.
7. If the problem still happens, **then check the GPU and storage** next.

Conclusion: I would start with RAM because a loose or faulty RAM module is a common and simple hardware cause of random restarts.

